Concepts

Astra Control Service

NetApp
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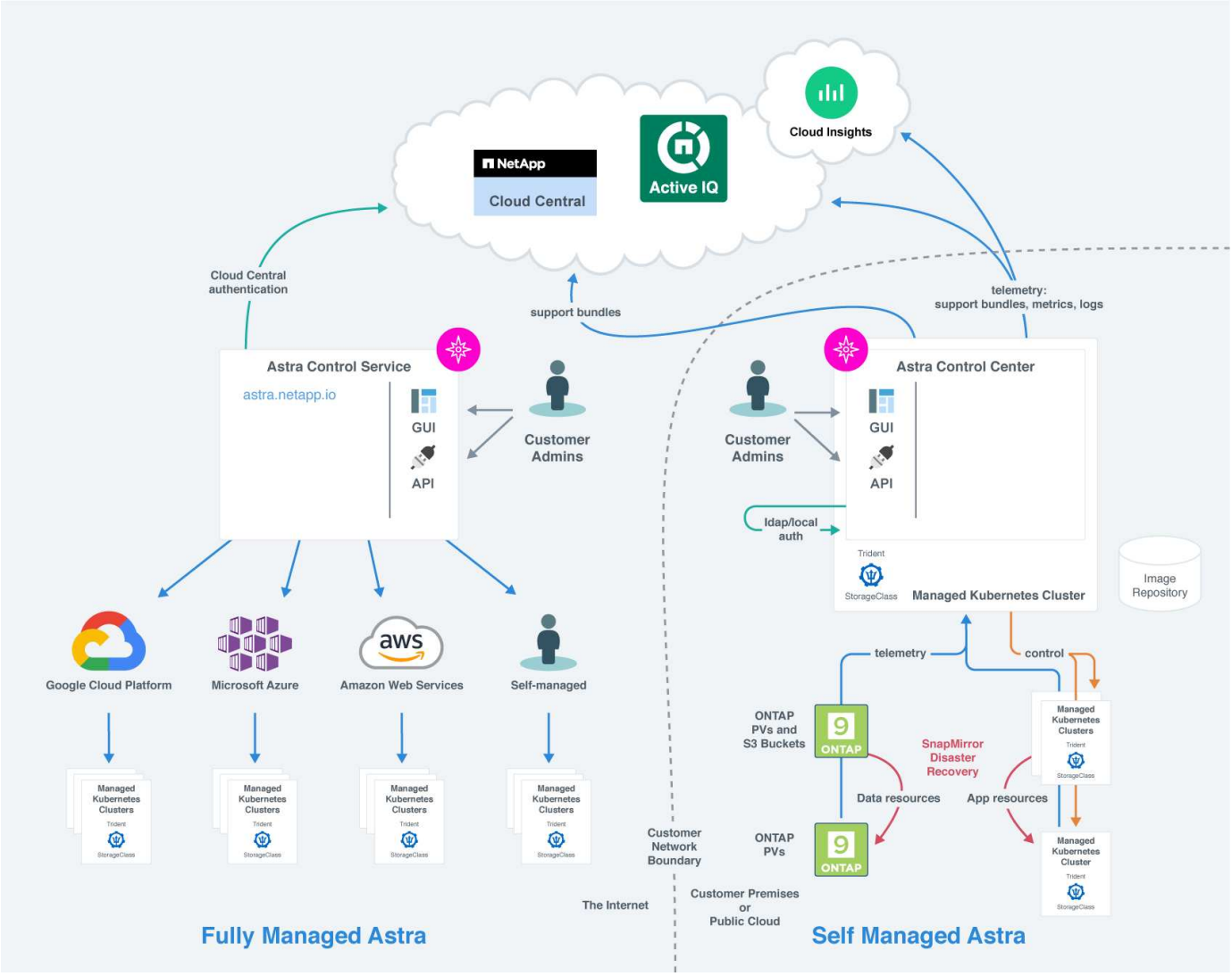
Concepts

Architecture and components

Here is an overview of the various components of the Astra Control environment.



Astra Control components

- **Kubernetes clusters**: Kubernetes is a portable, extensible, open-source platform for managing containerized workloads and services, that facilitates both declarative configuration and automation. Astra provides management services for applications hosted in a Kubernetes cluster.
- **Astra Control Provisioner**: As a fully supported storage provisioner and orchestrator maintained by NetApp that contains Astra Trident CSI functionality along with extended storage management capabilities, Astra Control Provisioner enables you to create storage volumes for containerized applications managed by Docker and Kubernetes. When deployed with Astra Control, Astra Control Provisioner includes a configured ONTAP storage backend.
- **Cloud Insights**: A NetApp cloud infrastructure monitoring tool, Cloud Insights enables you to monitor performance and utilization for your Kubernetes clusters managed by Astra Control Center. Cloud Insights

correlates storage usage to workloads. When you enable the Cloud Insights connection in Astra Control Center, telemetry information shows in Astra Control Center UI pages.

Astra Control interfaces

You can complete tasks using different interfaces:

- **Web user interface (UI)**: Both Astra Control Service and Astra Control Center use the same web-based UI where you can manage, migrate and protect apps. Use the UI also to manage user accounts and configuration settings.
- **API**: Both Astra Control Service and Astra Control Center use the same Astra Control API. Using the API, you can perform the same tasks that you would using the UI.

Astra Control Center also enables you to manage, migrate, and protect Kubernetes clusters running within VM environments.

For more information

- Astra Control Center documentation
- Astra Control API documentation
- Astra Trident documentation
- Cloud Insights documentation
- ONTAP documentation

Data protection

Learn about the available types of data protection in Astra Control Service, and how best to use them to protect your apps.

Snapshots, backups, and protection policies

Both snapshots and backups protect the following types of data:

- The application itself
- Any persistent data volumes associated with the application
- Any resource artifacts belonging to the application

A *snapshot* is a point-in-time copy of an app that's stored on the same provisioned volume as the app. They are usually fast. You can use local snapshots to restore the application to an earlier point in time. Snapshots are useful for fast clones; snapshots include all of the Kubernetes objects for the app, including configuration files. Snapshots are useful for cloning or restoring an app within the same cluster.

A *backup* is based on a snapshot. It is stored in the external object store, and because of this, can be slower to take compared to local snapshots. You can restore an app backup to the same cluster, or you can migrate an app by restoring its backup to a different cluster. You can also choose a longer retention period for backups. Because they are stored in the external object store, backups generally offer you better protection than snapshots in cases of server failure or data loss.

A *protection policy* is a way to protect an app by automatically creating snapshots, backups, or both according to a schedule that you define for that app. A protection policy also enables you to choose how many snapshots

and backups to retain in the schedule, and set different schedule granularity levels. Automating your backups and snapshots with a protection policy is the best way to ensure each app is protected according to the needs of your organization and service level agreement (SLA) requirements.

> *You can't be fully protected until you have a recent backup*. This is important because backups are stored in an object store away from the persistent volumes. If a failure or accident wipes out the cluster and its associated persistent storage, then you need a backup to recover. A snapshot would not enable you to recover.

> If you perform a snapshot or backup, but the operation fails with the error "The resource wasn't created because of an internal server issue", check to make sure the storage backend you are using has the correct drivers installed. Some storage backends need Container Storage Interface (CSI) drivers, while others need an external snapshot controller.

Immutable backups

An immutable backup is a backup that cannot be changed or deleted during a specified period. When you create an immutable backup, Astra Control checks to ensure that the bucket you are using is a write once read many (WORM) bucket, and if so, ensures that the backup is immutable from within Astra Control.
Astra Control Service supports creating immutable backups with the following platforms and bucket types:

- Amazon Web Services using an Amazon S3 bucket with S3 Object Lock configured
- Microsoft Azure using an Azure bucket with a retention policy configured
- Google Kubernetes Engine (GKE) using a Google Cloud Storage bucket with a retention policy configured
- NetApp StorageGRID using an S3 bucket with S3 Object Lock configured

Note the following when working with immutable backups:

- If you back up to a WORM bucket in an unsupported platform or to an unsupported bucket type, you might get unpredictable results, such as backup deletion failing even if the retention time has elapsed.
- Astra Control does not support data lifecycle management policies or manual deletion of objects on the buckets you use with immutable backups. Make sure that your storage backend is not configured to manage the lifecycle of Astra Control snapshots or backed up data.

Clones

A *clone* is an exact duplicate of an app, its configuration, and its persistent data volumes. You can manually create a clone on either the same Kubernetes cluster or on another cluster. Cloning an app can be useful if you need to move applications and storage from one Kubernetes cluster to another.

Storage classes and PV size for AKS clusters

Astra Control Service supports Azure NetApp Files, Azure managed disks, or NetApp Cloud Volumes ONTAP as the storage backend for Azure Kubernetes Service (AKS) clusters.

Azure NetApp Files

Astra Control Service supports Azure NetApp Files as the storage backend for Azure Kubernetes Service (AKS) clusters. You should understand how choosing a storage class and persistent volume size can help you

meet your performance objectives.

Service levels and storage classes

Azure NetApp Files supports three service levels: Ultra storage, Premium storage, and Standard storage. Each of these service levels are designed for different performance needs:

Ultra storage
: Provides up to 128 MiB/s of throughput per 1 TiB.

Premium storage
: Provides up to 64 MiB/s of throughput per 1 TiB.

Standard storage
: Provides up to 16 Mib/s of throughput per 1 TiB.

These service levels are an attribute of a capacity pool. You need to set up a capacity pool for each service level that you want to use with your Kubernetes clusters. Learn how to set up capacity pools.

Astra Control Service uses these service levels as storage classes for your persistent volumes. When you add Kubernetes clusters to Astra Control Service, you're prompted to choose either Ultra, Premium, or Standard as the default storage class. The names of the storage classes are *netapp-anf-perf-ultra*, *netapp-anf-perf-premium*, and *netapp-anf-perf-standard*.

Learn more about these service levels in the Azure NetApp Files docs.

Persistent volume size and performance

As described above, the throughput for each service level is per 1 TiB of provisioned capacity. That means larger volumes provide better performance. So you should take both capacity and performance needs into consideration when provisioning volumes.

Minimum volume size

Astra Control Service provisions persistent volumes using a minimum volume size of 100 GiB, even if the PVC asks for a smaller volume size. For example, if the PVC in a Helm chart asks for 6 GiB, Astra Control Service automatically provisions a 100 GiB volume.

Application backups

If you back up an application that resides on Azure NetApp Files storage, Astra Control Service automatically temporarily expands the capacity pool. After the backup is complete, Astra Control Service shrinks the capacity pool to its previous size. Depending on your Azure subscription, you might incur storage charges when this happens. You can see a history of capacity pool resize events in the **Activity** page event log.

If the capacity pool exceeds the maximum size allowed by the Azure subscription during the resize operation, the backup operation fails, and a warning is triggered from the Azure API.

Azure managed disks

Astra Control Service can use Container Storage Interface (CSI) drivers to interface with Azure Managed Disks as a storage backend. This service provides block-level storage that is managed by Azure.

Learn more about Azure managed disks.

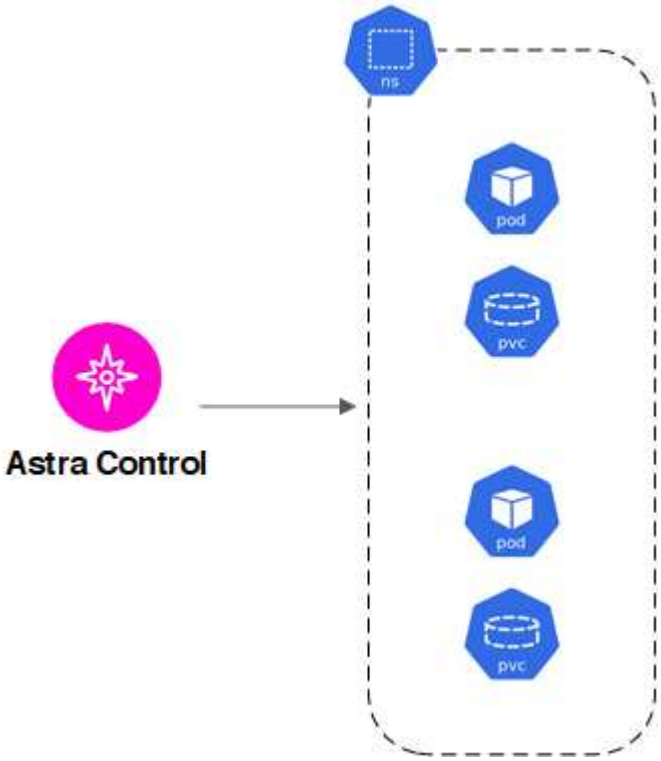
NetApp Cloud Volumes ONTAP

For specific information on configuring NetApp Cloud Volumes ONTAP, including performance recommendations, visit the NetApp Cloud Volumes ONTAP documentation.

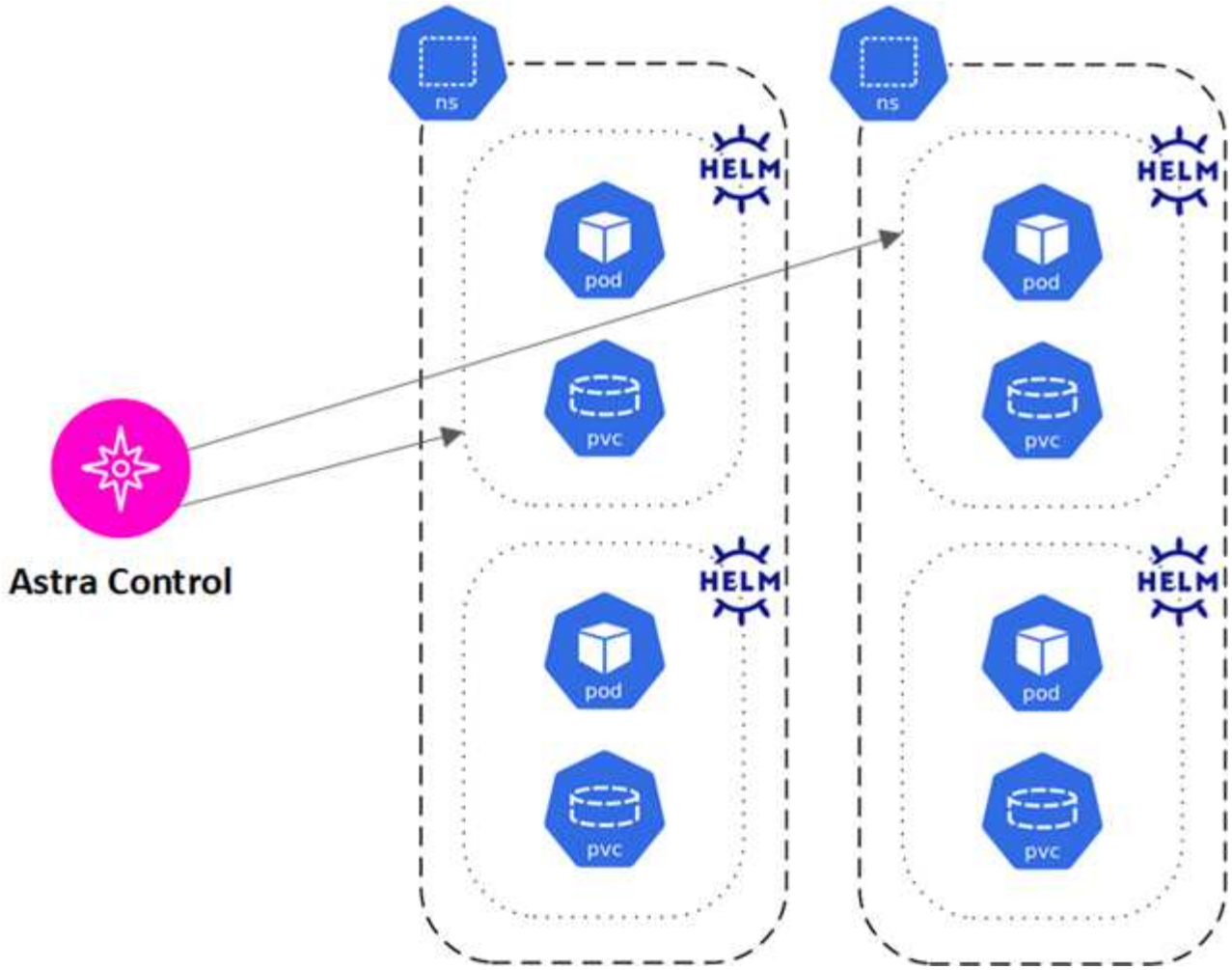
App management

When Astra Control discovers your clusters, the apps on those clusters are unmanaged until you choose how you want to manage them. A managed application in Astra Control can be any of the following:

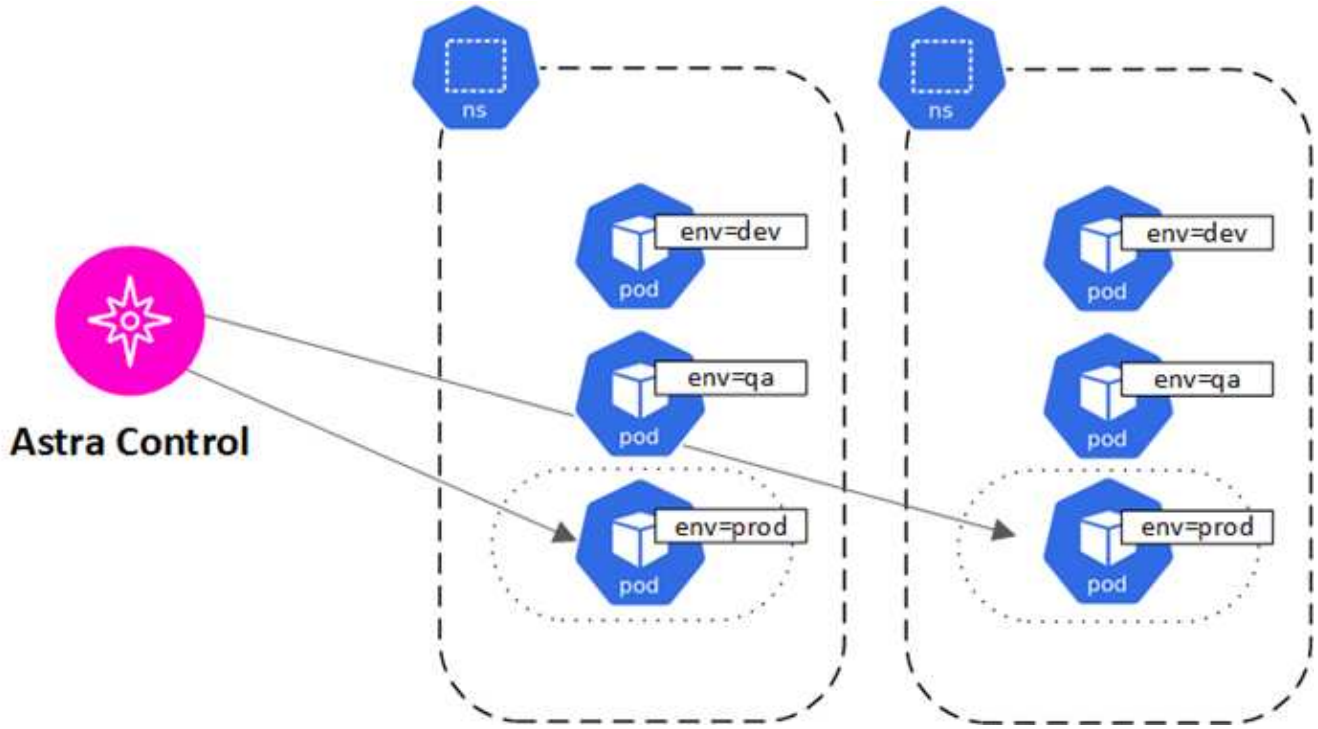
- A namespace, including all resources in that namespace



- An individual application deployed within one or more namespaces (Helm 3 is used in this example)



- A group of resources that are identified by a Kubernetes label within one or more namespaces



User roles and namespaces

Learn about user roles and namespaces in Astra Control, and how you can use them to control access to resources in your organization.

User roles

You can use roles to control the access users have to resources or capabilities of Astra Control. The following are the user roles in Astra Control:

- An **Owner** has Admin permissions and can delete accounts.
- An **Admin** has Member permissions and can invite other users.
- A **Member** can fully manage apps and clusters.
- A **Viewer** can view resources.

You can add constraints to a Member or Viewer user to restrict the user to one or more Namespaces.

Namespaces

A namespace is a scope you can assign to specific resources within a cluster that is managed by Astra Control. Astra Control discovers a cluster's namespaces when you add the cluster to Astra Control. Once discovered, the namespaces are available to assign as constraints to users. Only members that have access to that namespace are able to use that resource. You can use namespaces to control access to resources using a paradigm that makes sense for your organization; for example, by physical regions or divisions within a company. When you add constraints to a user, you can configure that user to have access to all namespaces or only a specific set of namespaces. You can also assign namespace constraints using namespace labels.

Find more information

- Manage roles

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